

PULSEGRID • IERBMS DEFENSE DOSSIER

Intelligent Emergency Response & Bed Management System • Final Examination Guide

Project Scope: Intelligent Geospatial Triage & Dynamic Bed Allocation	Defense Date: September 2026
Target Domain: Ghana Health Service (GHS) & National Ambulance Service (NAS)	Architecture: Microservices + Serverless Spatial PostgreSQL

1. Team Roles & Technical Division of Labor

Examiners rigorously evaluate individual ownership to ensure all team members deeply understand their technical deliverables. Below is the official division of engineering responsibilities:

Team Member	Official Role	Primary Deliverables & Technologies	Key Defense Focus
Larbi	Backend & Database Engineer	Node.js, Express, Socket.IO, Neon PostgreSQL 16, PostGIS 3.4, Bed Lifecycle Service (bedLifecycle.js), CARTO LDS Router, JWT / RBAC	Atomic transactions, bed reservation state machine, spatial queries, WebSocket event mesh
Kobby	Frontend & Geospatial Engineer	React 18, Vite, TypeScript, Leaflet / React-Leaflet, OSRM Routing, Dual-Buffer Triage Engine (<2ms), Web Audio Telemetry HUD, IndexedDB PWA	Sub-second clinical matching, zero-key mapping fallbacks, real-time client state, audio queue deduplication
Abigail	UI/UX & Clinical Ergonomics Designer	Design System (Tailwind CSS v4 Tokens), Paramedic Cockpit HUD, SATS / TEWS Clinical Color Hierarchy, 2.5D Hospital Bed Capacity Mesh, Defense Tour Bar	Gloved touch targets (48px), high-contrast daylight/night readability, cognitive load reduction, ISO usability

1.1 Larbi — Backend & Database Engineering

Architectural Responsibility: Engineered server foundations, real-time telemetry pipelines, and spatial data persistence.

- **Atomic Bed Lifecycle Engine (bedLifecycle.js):** Overcame the critical 'arrival race condition' by implementing transactional bed reservation at the instant of dispatch (reserved_general_beds, reserved_icu_beds) rather than upon physical ER arrival.
- **PostGIS Geospatial Database:** Structured PostgreSQL on Neon Serverless, integrating PostGIS 3.4 with 2,500+ HOTOSM-verified Ghanaian health facilities and spatial GIST indexing for sub-18ms facility radius queries.
- **Socket.IO Telemetry Mesh:** Created event-driven broadcasting for live ambulance location updates, capacity broadcasts, and automated hospital diversion triggers.
- **Security & RBAC:** Implemented JWT authentication, bcrypt password hashing, and role-based access control protecting Paramedic, Doctor, Hospital Admin, and Authority endpoints.

Examiner Defense Q&A; for Larbi:

Q: 'What happens if two ambulances compete for the last available ICU bed simultaneously?'

A: 'We enforce atomic row-level locking transactions inside bedLifecycle.js. When a dispatch is confirmed, PostgreSQL executes an atomic decrement on available capacity and increments reserved_icu_beds within a single transaction. If available beds reach zero, the constraint fails immediately, preventing double-allocation and prompting the AI to route to the next ranked facility.'

Q: 'Why maintain both a Node.js API Gateway and a Python FastAPI service?'

A: 'Separation of Concerns. Node.js with Socket.IO provides an asynchronous, non-blocking I/O event loop ideal for handling thousands of continuous WebSocket GPS telemetry pings. Python FastAPI is used strictly for compute-heavy Random Forest model inference and matrix operations. Separating them ensures analytical machine learning never blocks real-time emergency dispatching.'

1.2 Kobby — Frontend & Geospatial Engineering

Architectural Responsibility: Built the single-page application, interactive routing canvas, client-side triage algorithms, and offline synchronization queue.

- **Dual-Buffer Clinical Recommendation Pipeline:** Eliminated UI freezing by decoupling clinical heuristics from external network calls. Phase 1 computes SATS/TEWS multi-criteria matching in browser memory in < 2ms; Phase 2 asynchronously enriches live traffic in the background.
- **Geospatial Mapping Engine (LeafletMap.tsx):** Integrated Leaflet with zero-watermark OpenStreetMap tiles, high-contrast dark tactical CSS filters, OSRM turn-by-turn road snapping, vehicle follow camera, and CARTO 15-minute TravelTime isochrones.
- **Hands-Free Audio Telemetry HUD (audioTelemetry.ts):** Implemented Web Speech API telemetry with queue-flushing and an 8-second deduplication window to guarantee instructions never loop or stutter.
- **Offline-First Resilience:** Configured IndexedDB / localStorage queue that caches emergency intakes during network blackouts and auto-syncs via WebSockets upon signal restoration.

Examiner Defense Q&A; for Kobby:

Q: 'How do you guarantee that paramedics are not stranded if cellular network drops in a rural transit corridor?'

A: 'The frontend operates as an offline-first Progressive Web App (PWA). Emergency intakes and vitals are committed locally to IndexedDB with a unique client-side UUID. Local client heuristics calculate immediate hospital recommendations from cached facility coordinates. As soon as cellular connectivity is restored, our Socket.IO replay mechanism synchronizes all pending intakes with the central database.'

1.3 Abigail — UI/UX & Clinical Ergonomics

Architectural Responsibility: Spearheaded user interface design, clinical color theory, ergonomic tablet layouts, and human factors compliance.

- **Clinical Triage Hierarchy (SATS / TEWS):** Designed user interfaces adhering to the South African Triage Scale endorsed by Ghana Health Service: Red (Resuscitation/Priority 1), Orange (Very Urgent/Priority 2), Yellow (Urgent), and Green (Non-urgent).
- **Paramedic Cockpit HUD Ergonomics:** Designed oversized, high-contrast turn maneuver icons and ETA badges with minimum 48x48px touch targets, allowing paramedics wearing sterile nitrile gloves to operate tablets in vibrating ambulances without misclicks.
- **Dynamic Theme Token Architecture (theme.css):** Engineered consolidated Tailwind CSS v4 design tokens ensuring Light Mode cards evaluate to crisp white (#ffffff) with high-contrast text (#0f172a) and Dark Mode evaluates to deep tactical navy (#111c2d).
- **2.5D Hospital Capacity Mesh & Examiner Tour Bar:** Crafted visual stacked-bar representations for general vs. ICU bed capacities and designed the floating Defense Demo Bar for seamless 1-click presentation workflows.

Examiner Defense Q&A; for Abigail:

Q: 'What ergonomic principles guided the design of the in-transit ambulance cockpit?'

A: 'I designed under ISO 9241-11 human-centered design principles for high-stress, high-vibration physical environments. Elements are placed in natural thumb-reach zones, touch targets exceed 48 pixels, typography utilizes tabular numerals to prevent layout shift during live speed updates, and high-contrast color coding communicates urgency without requiring dense reading.'

2. System Infrastructure & Deployment Topology

PULSEGRID utilizes a 5-tier hybrid cloud architecture balancing edge performance with centralized data integrity. Below is the technical network topology as formalized in **UML Deployment Diagram 07**:

Tier Layer	Node / Hardware	Software / Runtime	Protocols & Security
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Tier 1: Client Edge	Ambulance Tablet (Android/iOS), Hospital ER Console, Command Dashboard	Chromium Core PWA, React 18 SPA, IndexedDB Storage, HTML5 Geolocation	HTTPS / WSS (Port 443), Web Speech API, NMEA GPS
Tier 2: Edge CDN	Vercel Global Edge Network	Edge Cache, HTTP/3, Static Bundling, Automated CI/CD	TLS 1.3, DDoS Mitigation, Geo-DNS Routing
Tier 3: API Gateway	Application Cluster (Node.js Container)	Node.js 20 LTS, Express Router, Socket.IO Engine, bedLifecycle.js	JWT Verification, CORS, Rate Limiting, WSS Fan-out
Tier 4: ML Engine	Predictive Analytics Microservice Container	Python 3.11, FastAPI ASGI (Uvicorn), Scikit-Learn Random Forest Pipeline	Internal HTTP REST (Port 8000), JSON Payloads
Tier 5: Database	Neon Serverless Spatial PostgreSQL Cluster	PostgreSQL 16 Engine, PostGIS 3.4 Spatial Extensions, 2,500+ Facilities	Encrypted TCP/SSL (Port 5432), GIST Spatial Indices

3. Detailed Guide to All 7 UML Diagrams

All visual diagrams are archived in the docs/uml/ directory. Use the explanations below when examiners ask you to walk through specific diagrams:

3.1 UML Use Case Diagram (01_uml_use_case_diagram)

Purpose: Establishes system functional boundaries and actor interactions.

- **Actors:** 5 Human Actors (Paramedic/EMS, ER Doctor, Ward Nurse, Command Dispatcher, Health Authority) and 1 System Actor (AI & Routing Engine).
- **Crucial Relationships:** UC-2 (Intake Vitals) has an <<include>> relationship to UC-3 (Compute SATS/TEWS) and UC-4 (Evaluate Hard Safety Constraints). UC-2 has an <<extend>> relationship to UC-12 (Enqueue Offline Payload) conditioned on network failure.

3.2 UML Class Diagram (02_uml_class_diagram)

Purpose: Represents the static domain model, relationships, and object methods.

- **Core Classes:** User (RBAC, JWT token generation), EmergencyCase (status management, bed assignment), PatientVitals (TEWS calculation, Glasgow Coma Scale derivation), Hospital (bed allocation, occupancy metrics), and Ambulance (telemetry updates, dispatch status).
- **Cardinality:** 1 Hospital aggregates 1..* Ambulances; 1 EmergencyCase is associated with exactly 1 PatientVitals record and 1 assigned Hospital.

3.3 UML Sequence Diagrams (03a, 03b, 03c)

Purpose: Illustrates runtime chronological message exchanges across subsystems.

- **03a Intake & Triage:** Paramedic inputs vitals $\rightarrow$ Frontend triggers instant client triage $\rightarrow$ Async call to FastAPI Random Forest model $\rightarrow$ Ranked hospital cards rendered.
- **03b Bed Lock Fan-Out:** Paramedic confirms dispatch $\rightarrow$ Backend executes atomic update in bedLifecycle.js $\rightarrow$ PostgreSQL locks bed $\rightarrow$ Socket.IO broadcasts capacity update to all hospital and dispatcher screens.
- **03c Offline Deadzone Sync:** Connectivity severed $\rightarrow$ PWA commits payload to IndexedDB $\rightarrow$ Connection restored $\rightarrow$ WebSocket handshake automatically pushes queued mutations to database.

3.4 UML Activity Diagram with Swimlanes (04_uml_activity_diagram_swimlanes)

Purpose: Outlines dynamic end-to-end operational workflow across 4 organizational swimlanes.

- **Swimlanes:** 1) Paramedic / Field Unit, 2) API Gateway & Socket Mesh, 3) AI Clinical Scoring Engine, 4) Hospital ER Triage.
- **Critical Decision Node:** Evaluates Glasgow Coma Scale (GCS < 8) or Polytrauma. If affirmative, execution path branches directly to ICU bed reservation with immediate trauma bay surgical prep alerts.

3.5 UML State Machine Diagram (05_uml_state_machine_diagram)

Purpose: Models the discrete lifecycle states of an emergency case and bed allocations.

- **State Progression:** Reported $\rightarrow$ Triage $\rightarrow$ Dispatched (Bed Reserved) $\rightarrow$ In-Transit $\rightarrow$ Arrived (Bed Occupied) $\rightarrow$ Treated / Discharged (Bed Freed).
- **Exception States:** Handles Diverted (rerouting to secondary hospital) and Cancelled, both of which automatically restore the reserved bed counter to the pool.

3.6 UML Component Diagram (06_uml_component_diagram)

Purpose: Visualizes high-level modular dependencies and decoupled subsystems.

- **Frontend Modules:** Navigation Engine, Triage Evaluator, RealTimeProvider, Voice HUD.
- **Backend Modules:** Auth Controller, Case Controller, Bed Lifecycle Service, Socket Mesh.
- **Analytics Modules:** Feature Extractor, Random Forest Predictor, Routing Heuristic Engine.

3.7 UML Deployment Diagram (07_uml_deployment_diagram)

Purpose: Maps logical software artifacts to physical execution hardware, cloud containers, and network ports (detailed in Section 2 above).

4. The 5-Minute 'Golden Defense' Presentation Script

Execute this structured 5-step sequence during your presentation to demonstrate the entire system smoothly in under 5 minutes without getting trapped:

Step 1: Automated Scenario Injection (Lead: Abigail / Kobby)

Click 'Examiner Defense Tour' in the top-right header and select 'Scenario 1: High Acuity Trauma'. Explain that the system loads realistic Ghana Health Service emergency cases rather than manual dummy inputs.

Step 2: Instant Dual-Buffer AI Matching (Lead: Kobby)

Click 'Calculate AI Hospital Recommendations'. Point out that ranked facilities load instantly (< 2ms) using local clinical heuristics while background ML refines traffic with an AbortController timeout.

Step 3: Cockpit Navigation & 15-Min Reach (Lead: Kobby / Abigail)

Confirm dispatch to launch the cockpit HUD. Demonstrate turn-by-turn guidance, vehicle chase camera ('Follow'), and toggle '15-Min Catchment' to show the CARTO TravelTime golden-hour polygon.

Step 4: Real-Time Bed Reservation (Lead: Larbi)

Switch to Hospital Dashboard using the role switcher. Show the 'Incoming Reserved' indicator on the bed capacity bar. Explain how bedLifecycle.js prevents arrival race conditions.

Step 5: Offline-First Demonstration (Lead: Kobby / Larbi)

Simulate network loss (Offline in DevTools). Submit an intake to show it queued in IndexedDB. Reconnect network to watch Socket.IO auto-sync and update the central database.

Conclusion: This architectural blueprint and role matrix certifies that the PULSEGRID system is academically rigorous, clinically defensible, and fully aligned with Ghana Health Service operational standards.